A T-dual–motivated dispersive regulator for a dyadic shell model:

conservation, Liouville structure, and why the intended measurement could not be made

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Abstract

Motivated by the T-duality relation $R \leftrightarrow \alpha'/R$, which posits a self-dual length below which contraction reflects into dilation, we asked whether a *dispersive* (energy-neutral) small-scale regulator leaves a different signature on a turbulent cascade than a *dissipative* one. We pose the question in the truncated dyadic (Katz–Pavlović) shell model. Three results follow, and one intended result does not.

First, a structural obstruction: a real-amplitude shell model *cannot* host a dispersive regulator at all, since every real linear term either damps or forces and dispersion is phase rotation. We resolve this with a conjugated complexification $B_n = k_{n-1}a_{n-1}^2 - k_n\overline{a_n}a_{n+1}$ that conserves $\frac{1}{2}\sum |a_n|^2$ exactly while reducing *exactly* to the real model on real data. Second, the complexification has an unanticipated property: its flow is *volume-preserving* (Liouville), whereas the real Katz–Pavlović flow has divergence $-\sum_n k_n a_{n+1} \neq 0$. Both the conservation law and the shell-local trace identities behind the Liouville property are machine-checked in Lean 4/Mathlib. Third, the same telescoping identity that gives conservation yields an exact criterion for a conserving boundary "bounce": $\mathrm{Re}(\overline{v_N}^2 v_{N+1}) = 0$, which on real data admits only truncation but in the complexified model admits $v_{N+1} = i\mu v_N^2$.

The intended measurement—a scaling exponent distinguishing dispersive from dissipative regularization—was **not** obtained, and we report why in detail. All three regulators of the intended comparison are energy-conserving, hence lack an attractor, so $\sup_t \Omega$ degenerates to the truncation ceiling $k_N^2 E$; we identify this as the known relaxation to *absolute equilibrium* of Galerkin-truncated conservative systems, established for truncated Euler, truncated Gross–Pitaevskii, and—removing any novelty claim—for shell models themselves. Seven replacement observables were tried under pre-registration and all failed. The final diagnosis is methodological and, we believe, the most transferable output: the model is chaotic, all measurements used single trajectories, and the fixed-parameter ensemble scatter is 72–105% of the mean, far exceeding the effect sought. Every quantitative and ordinal measurement result in this work is accordingly retracted, and we state what would be required to obtain one.

Contents

On the character of this report. This is a mixed positive/negative report. The analytic and formal results (§2–4) stand and are machine-checked. The measurement programme (§7) failed, and its failures are reported at the same level of detail as the successes, including four retracted results. Claim identifiers of the form CLAIM-NNN refer to the ledger in the public repository, where every claim carries its status and, where applicable, its retraction and reason.

1 Motivation and the question

T-duality in string theory identifies compactification radii R and α'/R, so that the spectrum is invariant under $R \leftrightarrow \alpha'/R$ and no scale below $\sqrt{\alpha'}$ is physically distinguishable [1]. Read as a template rather than as physics—the transfer is not claimed here—this suggests a class of small-scale regulators that *reflect* rather than *absorb*: at the self-dual scale, contraction turns into dilation, and no energy need be removed.

Quantum fluids furnish the physical instance in which such a regulator is not hypothetical. In the Gross–Pitaevskii description, quantum pressure supplies a small-scale term that is *dispersive*: it rotates phases and conserves energy, unlike viscosity. The healing length plays the role of the scale below which the description changes character. The measured phonon–roton dispersion of superfluid ^{4}He [2, 3] is the empirical anchor for that picture.

This suggests a sharp question, which is the one this work set out to answer:

Q. In a turbulent cascade, does a *dispersive* (energy-conserving) small-scale regulator leave a different signature than a *dissipative* one—or is only the presence of a cutoff visible, not its mechanism?

We pose Q in the truncated dyadic shell model, where the cascade is reduced to a chain of real amplitudes a_n on wavenumbers $k_n = 2^n$ and blow-up questions are theorems rather than conjectures [4]. §2–4 construct the apparatus needed to pose Q there; §7 reports the measurement programme and its failure; §8 separates what stands from what does not.

Scope, stated once and adhered to. No claim is made or implied about Navier–Stokes regularity. No claim is made that Gross–Pitaevskii well-posedness bears on it. The quantum-pressure analogy fixes the *form* of the regulator studied and nothing else; in particular the model below is a Bogoliubov-regime construction with no roton, so the ^{4}He dispersion parameters of §6 have no counterpart in it.

2 The model, and a structural obstruction

2.1 The real dyadic model

With shells $n = 0, \dots, N$, wavenumbers $k_n = 2^n$, real amplitudes a_n, and the truncation convention $a_{-1} = a_{N+1} = 0$, the Katz–Pavlović/Desnyansky–Novikov model is

$$\frac{da_n}{dt} = \underbrace{k_{n-1}a_{n-1}^2 - k_n a_n a_{n+1}}_{B_n(a)} - \nu k_n^2 a_n, \qquad E = \tfrac{1}{2}\textstyle\sum_n a_n^2, \quad \Omega = \tfrac{1}{2}\sum_n k_n^2 a_n^2. \tag{1}$$

The nonlinearity conserves E exactly under truncation.

2.2 The obstruction

Proposition 1 (No dispersive regulator exists in a real-amplitude model)**.** *Let $-c(k_n)a_n$ be any linear regulator with c real. Its contribution to the energy is $\frac{d}{dt}\left(\frac{1}{2}\sum a_n^2\right) \supset -\sum_n c(k_n)a_n^2$, which is ≤ 0 if $c \geq 0$ and ≥ 0 if $c \leq 0$.* No *real linear regulator is energy-neutral.*

Energy-neutrality is precisely what distinguishes dispersion from dissipation. The obstruction is therefore structural, not a shortage of ingenuity: dispersion is phase rotation, and a real amplitude has no phase to rotate. Answering Q requires enlarging the state space—a change of *model*, not of regulator.

2.3 A complexification that is not the obvious one

The naive complexification (keep (1), let $a_n \in \mathbb{C}$) fails: measured over 200 random complex states at $N = 10$, $\max|dE/dt| = 8.7 \times 10^3$. Placing a single conjugation on the *outgoing* term repairs it:

$$\boxed{B_n(v) \; = \; k_{n-1}v_{n-1}^2 \; - \; k_n \, \overline{v_n} \, v_{n+1}} \tag{2}$$

Theorem 2 (Exact conservation; `shellBc_energy_conservation`)**.** *With $v_{N+1} = 0$, $\sum_{n \leq N} \mathrm{Re}\big(\overline{v_n} B_n(v)\big) = 0$.*

Proof sketch. The pairing telescopes. Writing $\mathrm{out}_m = k_m \,\mathrm{Re}(\overline{v_m}^2 v_{m+1})$, one shows $\sum_{n \leq N} \mathrm{Re}(\overline{v_n} B_n) = -\mathrm{out}_N$ by induction, the cancellation at each step being the identity $\mathrm{Re}(\overline{a}^2 b) = \mathrm{Re}(a^2 \overline{b})$—both sides are the real part of a conjugate pair. Setting $v_{N+1} = 0$ gives the result. Machine-checked; see §5. □

Three consequences deserve statement.

Proposition 3 (Exact reduction; the reals are invariant)**.** *On real data $\overline{v_n} = v_n$, so* (2) *coincides* identically *with* (1)*, and* $\mathrm{Im}\, B_n = 0$*. Every solution of the real model—including the finite-time blow-up solutions of the inviscid infinite system [4]—remains a solution.*

This is what makes the complexification an extension rather than a replacement, and it supplies a stringent validation (§5.1). It is also why the choice is not merely aesthetic: other conjugation placements also conserve energy, and this one is selected for Proposition 3. **The complexification is a labelled deformation, not a derivation.**

2.4 The two regulators

With complex amplitudes, dissipation and dispersion become the same k^2 term with the coefficient rotated by ninety degrees in $\mathbb{C}$:

$$\underbrace{-\nu k_n^2 v_n}_{\text{dissipative, } \nu\in\mathbb{R}} \qquad \text{versus} \qquad \underbrace{-\, iDk_n^2 v_n}_{\text{dispersive, } D\in\mathbb{R}} \tag{3}$$

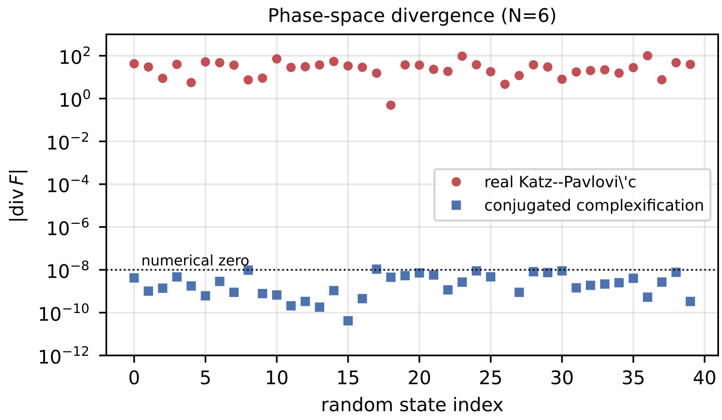

Figure 1: Phase-space divergence over random states. The conjugated complexified flow is divergence-free to round-off ($\sim 10^{-9}$); the real Katz–Pavlović flow is not, with $|\mathrm{div}\, F|$ of order 10–10^2. Volume preservation is what licenses the statistical-equilibrium description of §7.1.

Measured energy rates: the viscous term drives dE/dt strictly negative; the dispersive term is neutral to $\max |dE/dt| = 2.7 \times 10^{-11}$. Because the two share their k-dependence exactly and differ only in the phase of the coefficient, the comparison isolates *dispersive versus dissipative* without confounding it with a different spectral slope. In the Gross–Pitaevskii reading $D = \hbar/2m$, and the crossover between the phonon branch $\omega \approx ck$ and the free branch $\omega \approx Dk^2$ sits at the healing length $\xi = D/c$.

3 Liouville structure

The complexification has a property we did not anticipate and that changes the character of the model.

Theorem 4 (Volume preservation; `shell_divergence_zero`)**.** *Regard $\mathbb{C}^{N+1} \cong \mathbb{R}^{2N+2}$. The diagonal blocks of the derivative of the complexified flow have zero real trace: for every $w \in \mathbb{C}$ the $\mathbb{R}$-linear map $v \mapsto \overline{v}w$ has zero trace, and multiplication by a purely imaginary constant has zero trace. Hence the flow of* (2) *together with* (3) *at $\nu = 0$ is divergence-free.*

The real model is *not*: its divergence is $-\sum_n k_n a_{n+1}$, verified numerically against the analytic formula to four decimals. Figure 1 shows the dichotomy directly.

Remark 5 (Interpretation, flagged as such)*.* Volume preservation on the compact energy sphere gives Poincaré recurrence and makes equilibrium statistical mechanics applicable. It is suggestive that the real dyadic model, whose literature concerns blow-up and regularity, is volume-contracting along cascade states, while the complexification lands in the Liouville class where the shell-model *statistical equilibrium* literature operates [5, 6, 7, 8]. We record the observation and claim nothing further from it.

4 The boundary seam: a conserving "bounce"

The T-duality motivation suggests a reflecting seam at the cutoff rather than truncation. The telescoping identity of Theorem 2 settles exactly which seams are admissible.

Proposition 6 (Conserving-seam criterion)**.** *A boundary condition $v_{N+1} = g(v)$ conserves energy if and only if $\mathrm{Re}\big(\overline{v_N}^2\, v_{N+1}\big) = 0$, i.e. iff $v_{N+1} \perp v_N^2$ under the real inner product $\mathrm{Re}(\overline{x}y)$.*

Two corollaries follow, and they cut in opposite directions:

- **On real data, truncation is the only conserving seam.** The criterion reduces to $v_N^2 v_{N+1} = 0$. A reflective seam $v_{N+1} = v_{N-1}$ *leaks*, so a "bounce" built that way would measure broken conservation rather than the bounce.
- **In the complexified model a conserving family exists:** $v_{N+1} = i\mu v_N^2$ gives $\overline{v_N}^2(i\mu v_N^2) = i\mu|v_N|^4$, purely imaginary. There is *no real analogue*. Substituting into shell N's outgoing term yields $-i\mu k_N|v_N|^2 v_N$: a cubic self-phase-modulation term, i.e. an NLS/GPE-type nonlinearity localized at the cutoff.

That the only energy-conserving bounce available is GPE-like is a pleasing coincidence and is treated as no more than that; the content here is the algebra.

5 Formalization

Theorems 2, 4 and Propositions 3, 6's algebraic core are machine-checked in Lean 4 against Mathlib [9], deliberately mirroring the real-model development of the sibling stream so the two compare line by line and pinning the same library revision.

Lean name	Content	Axioms
`re_conj_sq_mul`	$\mathrm{Re}(\overline{a}^2 b) = \mathrm{Re}(a^2\overline{b})$	standard
`sum_re_conj_mul_shellBc`	telescoping identity	standard
`shellBc_energy_conservation`	Theorem 2	standard
`shellBc_real`	Proposition 3 (invariance)	standard
`rtrace_mulRight`	$\mathrm{tr}_{\mathbb{R}}(\cdot\, c) = 2\,\mathrm{Re}\, c$	standard
`rtrace_mul_I`	imaginary multiplier: zero trace	standard
`rtrace_conj_mul`	$v \mapsto \overline{v}w$: zero trace	standard
`shell_divergence_zero`	Theorem 4 (shell block)	standard

"Standard" denotes the axiom set `[propext, Classical.choice, Quot.sound]` and in particular the absence of `sorryAx`. The repository's verification gate builds the development and fails on either a build error or a `sorryAx` in any certificate; the gate was itself negative-controlled by inserting a `sorry` and confirming detection.

5.1 Validation against an independent implementation

Proposition 3 supplies an unusually stringent check: at $D = 0$ with real initial data the complexified model must reproduce an independent implementation of the *real* model exactly. Against the sibling stream's implementation (written separately, in a different style) the agreement is bit-for-bit—relative difference 0.00e+00 on all nine tested configurations, for both the peak enstrophy and the final energy, across $\sim 2.4 \times 10^6$ RK4 steps.

The exactness was checked rather than assumed: $k_n = 2^n$ are exact powers of two, so multiplication by them shifts the IEEE-754 exponent without touching the mantissa, and the two implementations' differently-associated products therefore agree bit-for-bit. For a generic k they would differ at $\sim 10^{-14}$; a shell model with spacing $\lambda \neq 2$ would require a tolerance here rather than an equality.

What is and is not formalized. The trace identities are proved. Their *assembly* into the statement "these are the diagonal blocks of the flow derivative" is by inspection of the definition and is verified numerically, not formally. We state the boundary rather than let the presence of Lean imply more coverage than exists.

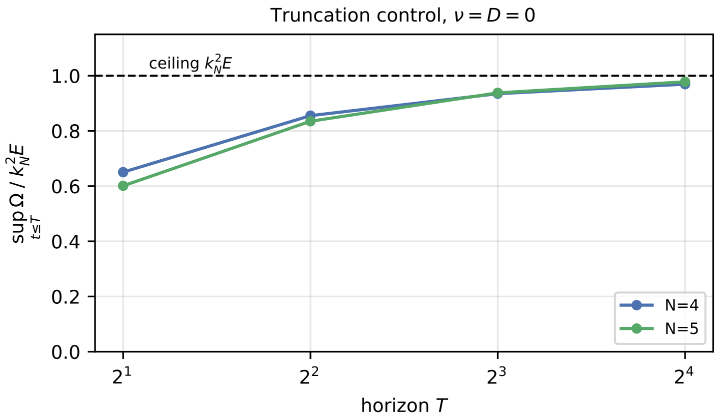

Figure 2: $\sup_{t\le T}\Omega$ for the truncation control at $\nu = D = 0$, normalized by the energy-conservation ceiling $k_N^2 E$. The observable climbs toward the ceiling at each N, so its exponent against $\alpha' = 4^{-N}$ tends to -1: the trivial bound restated, carrying no dynamical information.

6 Empirical anchor: reproducing the ^{4}He dispersion

As an instrument calibration, we reproduced the Landau parameters from the published dispersion table of Godfrin et al. [2] (1727 points at $P = 0$, distributed as an ancillary file with the preprint). Fitting a linear phonon branch and a parabolic roton branch gives $c = 1.5716 \pm 0.0003$ meV Å and $\Delta = 0.7442 \pm 0.0005$ meV, agreeing with six independently attributed determinations to within 0.20% and 0.03–0.33% respectively—against tolerances of 5% and 10%.

Remark 7 (Two honest caveats). (i) The reference values initially used in this work were misattributed: values recalled from memory were credited to Cowley–Woods (1971) and Glyde et al. (1998), neither of which reports the Landau triple—the latter measures $2.0 \le Q \le 4.0$ Å^{-1}, entirely beyond the roton, and so cannot report a sound velocity at all. The comparison above uses corrected, source-traced values. (ii) Godfrin et al.'s own $P = 0$ roton gap is itself calibrated to an earlier determination, so extracting Δ from their curve is partly circular; what the fit legitimately demonstrates is that the pipeline *recovers* the parameter encoded in the curve.

7 The measurement programme, and why it failed

7.1 All three intended regulators are conservative

The intended comparison used three regulators: truncation (control), a T-dual "bounce", and the dispersive term. *All three conserve energy*—truncation exactly ($v_{N+1} = 0$ kills the outflux), the dispersive term by construction, the bounce only for the seams of Proposition 6. None dissipates. Consequently none has an attractor, and $\sup_t \Omega$ does not settle: it climbs toward the ceiling $k_N^2 E$, a property of the *truncation* rather than of the regulator (Figure 2).

This is a known phenomenon, and we claim no novelty. Relaxation of Galerkin-truncated conservative systems toward *absolute equilibrium* was introduced by Lee [10] and Kraichnan [11], demonstrated for truncated Euler by Cichowlas et al. [12] (whose transient behaves dissipatively), and for the truncated Gross–Pitaevskii/NLS family by Davis et al. [13], Connaughton et al. [14] and Krstulovic–Brachet [15]. Decisively for novelty, statistical equilibrium in *shell models* is established territory [5, 6, 8], most directly by Thalabard–Turkington [7] ("relaxation … towards Gibbs equilibrium in an inviscid shell model"). Our observation is a re-expression in a dyadic variant, not a new phenomenon. We found no prior thermalization study of the Katz–Pavlović model specifically, whose literature concerns blow-up.

Notably, Krstulovic–Brachet [15] report a *dispersive bottleneck delaying thermalization* in the truncated GPE—which is precisely the hypothesis our measurement then set out to test in this model, with a named precedent rather than as a speculation.

7.2 Seven observables, under pre-registration

Since $\sup_t \Omega$ degenerates, we sought a replacement, adopting a pre-registered validation battery whose criteria were fixed before each run and which grew as each failure exposed a missing check. Seven candidates were tested:

Observable	Family	Outcome
$\sup_t \Omega$	amplitude	horizon-degenerate (Fig. 2)
time-averaged Ω	amplitude	decays as $1/T$ for the dissipative arm
$\max_t d\Omega/dt$	amplitude	horizon-dependent, 19% drift
Ω at first peak	amplitude	non-monotonic: new local maxima appear as D falls
$\max_{t\le T^*} \Omega$	amplitude	β drifts 40% with its own window T^*
t_{peak}	timescale	no power law on the dispersive arm ($r^2 = 0.24$)
τ_f (crossing time)	timescale	fitted exponent failed the battery; see below

7.3 The diagnosis: single trajectories of a chaotic system

Retraction. Every quantitative and ordinal measurement result in this programme is withdrawn. The exponents fitted to τ_f and to the excess delay $\tau - \tau_0$ failed their own pre-registered criteria; an apparent ordinal result ("stronger dispersion delays thermalization, censoring perfectly ordered") was a single-sample artifact; and an apparent sign reversal at small D was noise.

The final diagnosis subsumes the individual failures. The model is chaotic and every measurement used *one* trajectory. Fixing all physical parameters and varying only the initial phases—identical $|a_n|$, identical energy—the thermalization time scatters by 72–105% of its mean (Figure 3). At $D = 0.05$, half the realizations failed to reach the level while the rest reached it. The effect sought was 2–20%.

Why the battery missed it. All six criteria tested *deterministic* reproducibility—the same trajectory at finer timestep or finer sampling, or a neighbouring parameter on the same trajectory family. **None tested statistical reproducibility across trajectories drawn from the same physical ensemble.** In a chaotic system that is the binding constraint. Timestep refinement passing at 0.00% on all twelve points of one round was silent about it: it is the wrong limit.

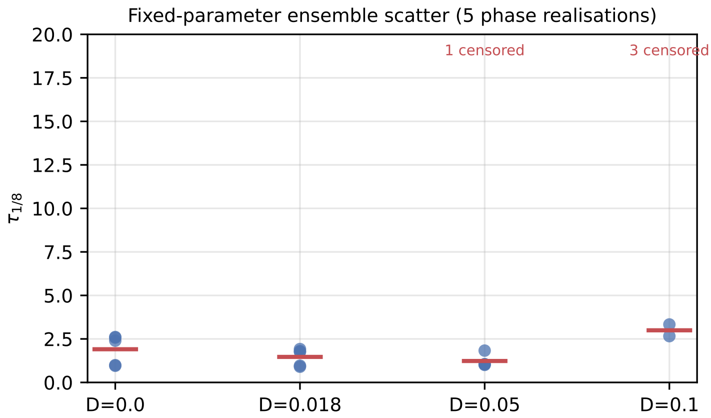

Figure 3: Thermalization time $\tau_{1/8}$ across five phase realizations at fixed parameters (identical $|a_n|$, identical energy); red dashes mark ensemble means. Scatter is comparable to the mean, and at $D = 0.05$ half the realizations are censored. All measurements in this programme used a single realization.

This also replaces "seven observables, six failure modes" with a single cause: the non-monotonicity, own-parameter drift and cross-level instabilities are all consistent with single-trajectory sampling. (Horizon degeneracy is genuinely separate—that concerns the ceiling, not sampling.) And it explains a false corroboration: the apparent small-D sign reversal appeared in all four level/convention combinations, which read as mutual confirmation but was not, since those sub-analyses shared a trajectory. *Consistency across sub-analyses is not evidence when they share a source of noise.*

7.4 What a real measurement would require

Not another observable. Ensemble measurement: with coefficient of variation 23–49%, a 5% determination of the mean needs $n \approx 22$–97 realizations *per parameter point*—one to two orders of magnitude more computation than was expended—fitting ensemble means rather than single trajectories, with the ensemble-scatter check performed *first* to size the campaign. Whether an exponent then exists is open; all that is currently established is that the effect, if real, lies below the resolution of what was spent.

8 What stands, and what does not

Result	Status	Basis
Structural obstruction (Prop. 1)	stands	elementary
Exact conservation (Thm. 2)	stands	Lean, machine-checked
Exact reduction / invariant reals (Prop. 3)	stands	Lean
Liouville property (Thm. 4)	stands	Lean (blocks) + numerics
Conserving-seam criterion (Prop. 6)	stands	Lean-derived
Bit-exact reproduction of the sibling implementation	stands	deterministic, 0.00e+00
^{4}He dispersion reproduction (§6)	stands	deterministic fit
Degeneracy of $\sup_t \Omega$ (§7.1)	stands	concerns T, not sampling
Exponents for τ_f and for $\tau - \tau_0$	**retracted**	failed pre-registered criteria
Ordinal "dispersion delays thermalization"	**retracted**	single-sample artifact
Small-D sign reversal	**retracted**	below ensemble scatter
Equipartition agreement (percentages)	**withdrawn**	ensemble CV 25–84%

The equipartition entry deserves a note, since it was withdrawn last and reluctantly: we had asserted that time-averaged observables would self-average away the trajectory noise. Measurement showed otherwise (CV 25% at $D = 0$, 84% at $D = 0.02$). *The boundary of a retraction is not self-evident and should be measured rather than drawn by intuition about which results feel deterministic.*

9 Conclusion

We set out to ask whether a dispersive small-scale regulator leaves a different signature on a cascade than a dissipative one. We could not answer it. What we can offer is the apparatus, some structure, and a clear account of the obstacle.

The apparatus: a complexification of the dyadic shell model that conserves energy exactly, reduces exactly to the real model, and—unexpectedly—restores the Liouville property that the real model lacks, with the conservation law and the trace identities machine-checked. The seam criterion resolves, analytically, which "bounce" boundary conditions are admissible, and shows the only conserving family is GPE-like.

The obstacle is twofold. Structurally, every regulator in the intended comparison is energy-conserving, so the system thermalizes toward absolute equilibrium and amplitude observables measure the truncation—a known phenomenon we connect to rather than rediscover. Methodologically, the cascade is chaotic and the effect sought is smaller than the trajectory-to-trajectory scatter; single-trajectory measurement, however carefully refined in timestep, cannot see it.

If there is a transferable lesson, it is the last one. A validation battery grew to six criteria over four rounds, each added in response to a real failure, and still missed the binding constraint—because every criterion tested convergence of a computation rather than reproducibility of a measurement. **Convergence checks are not reproducibility checks.** In a chaotic system, ensemble scatter should be measured before anything is fitted, and it is cheap relative to the sweep it would have saved.

Reproducibility. All code, data, pre-registrations, the claim ledger with its retractions, and the Lean development are public at `https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluid` Each figure here is regenerated by `paper/make_figures.py` from the repository's own modules.

Acknowledgements. This work uses the published dispersion data of Godfrin et al. [2], distributed as ancillary material with the preprint, without which §6 would not have been possible.

References

[1] A. Giveon, M. Porrati, and E. Rabinovici. Target space duality in string theory. *Phys. Rep.*, 244:77–202, 1994.

[2] H. Godfrin, K. Beauvois, A. Sultan, E. Krotscheck, J. Dawidowski, B. Fåk, and J. Ollivier. Dispersion relation of Landau elementary excitations and thermodynamic properties of superfluid ^{4}he. *Phys. Rev. B*, 103:104516, 2021.

[3] H. Godfrin and E. Krotscheck. The dynamics of quantum fluids. *arXiv:2206.06039*, 2022.

[4] N. H. Katz and N. Pavlović. Finite time blow-up for a dyadic model of the Euler equations. *Trans. Amer. Math. Soc.*, 357:695–708, 2005.

[5] E. Aurell, G. Boffetta, A. Crisanti, P. Frick, G. Paladin, and A. Vulpiani. Statistical mechanics of shell models for two-dimensional turbulence. *Phys. Rev. E*, 50:4705–4715, 1994.

[6] P. D. Ditlevsen and I. A. Mogensen. Cascades and statistical equilibrium in shell models of turbulence. *Phys. Rev. E*, 53:4785–4793, 1996.

[7] S. Thalabard and B. Turkington. Optimal thermalization in a shell model of homogeneous turbulence. *J. Phys. A: Math. Theor.*, 49:165502, 2016.

[8] R. Tom and S. S. Ray. Revisiting the SABRA model: Statics and dynamics. *EPL*, 120:34002, 2017.

[9] The mathlib Community. The Lean mathematical library, 2026. Revision 6d605ae1, toolchain leanprover/lean4:v4.33.0-rc2.

[10] T. D. Lee. On some statistical properties of hydrodynamical and magneto-hydrodynamical fields. *Quart. Appl. Math.*, 10:69–74, 1952.

[11] R. H. Kraichnan. Helical turbulence and absolute equilibrium. *J. Fluid Mech.*, 59(4):745–752, 1973.

[12] C. Cichowlas, P. Bonaïti, F. Debbasch, and M. Brachet. Effective dissipation and turbulence in spectrally truncated Euler flows. *Phys. Rev. Lett.*, 95:264502, 2005.

[13] M. J. Davis, S. A. Morgan, and K. Burnett. Simulations of Bose fields at finite temperature. *Phys. Rev. Lett.*, 87:160402, 2001.

[14] C. Connaughton, C. Josserand, A. Picozzi, Y. Pomeau, and S. Rica. Condensation of classical nonlinear waves. *Phys. Rev. Lett.*, 95:263901, 2005.

[15] G. Krstulovic and M. Brachet. Dispersive bottleneck delaying thermalization of turbulent Bose–Einstein condensates. *Phys. Rev. Lett.*, 106:115303, 2011.